DIGITAL TWIN CHALLENGE 2026

RespiraTwin:

An Asthma Digital Twin for Early Warning

A simulation-based respiratory risk forecasting prototype

TEAM NAME

AuraTwin

COLLEGE

Adani University

MEMBERS

Dhruvi Shah

All patient, wearable, pollution and event data are simulated. Not for clinical use.

THE PROBLEM

Asthma changes between clinic visits

Day-to-day signals can shift while the care team is not observing the patient.

POLLUTION	SYMPTOMS	ADHERENCE
Exposure changes	**Breathing changes**	**Treatment patterns**

The gap

A clinic snapshot may miss changing patterns between visits.

Early warning could support earlier review of a changing pattern.

INDIA CONTEXT

Air pollution and respiratory health

12.9% of global asthma cases,
42.3% of global asthma deaths

India, GBD 2019 estimates (Lung India, 2022).

A patient-specific state, updated over time

Six steps from historical context to an explorable forecast.

01

Load EHR profile

History + baseline

02

Ingest hourly signals

Pollution · wearable · adherence · symptoms

03

Update twin state

Align and validate observations

04

Forecast 24h risk

Model estimates event probability

05

Explain the output

Signal families + history

06

Ask what-if questions

Hypothetical inputs only

State updates and what-if responses are recorded separately from original observations.

SYSTEM ARCHITECTURE

Two streams. One virtual patient. A traceable forecast.

Architecture at a glance

INPUT STREAMS

Synthetic EHR
Synthetic pollution
Simulated wearables, adherence and symptoms

FEATURE FUSION

Schema + timestamp checks
Past-looking features
Personal baseline

TWIN ENGINE

Current state
24h forecast
Reason codes + history

DOCTOR DASHBOARD

Risk forecast
Replay and early-warning view
What-if laboratory

EHR + pollution + wearables → feature fusion → twin engine (state, forecast, explanations, history) → doctor dashboard and what-if

All input data shown in this prototype is simulated.

Every stream is simulated

Designed for a reproducible prototype without real patient records.

STREAM	FIELDS / CONTENT	PROVENANCE
EHR profiles	History, severity, prior events, baseline	300 synthetic patients
Pollution	PM2.5 / PM10 and illustrative AQI	Synthetic urban Delhi-range series; not real measurements
Wearables	Respiratory rate, SpO_2, heart rate, HRV, sleep, steps	Simulated time series
Adherence + diary	Controller adherence, rescue inhaler use, symptoms	Simulated logs

Cohort: 300 patients × 60 days × hourly

ALL DATA SIMULATED; NO REAL PATIENT DATA USED

EVENT GENERATION

The model learns a simulator—not asthma biology

Outcome labels are generated by explicit assumptions.

HAZARD DRIVERS

- Pollution and asthma severity
- Prior exacerbations and sensitivity
- Adherence and hidden vulnerability

SIMULATED EVENT BEHAVIOR

6–24h	Prodrome: respiratory rate up, SpO_2 down, rescue use and symptoms up
30%	Silent episodes with weak or no prodrome
72h	Refractory period after an onset

1,398 episodes · Positive label rate: 7.77%

Assumptions for modelling—not clinical facts.

MODEL AND SAFEGUARDS

Constrained learning with explicit leakage checks

Model choices are documented and repeatable.

52

PAST-LOOKING FEATURES

No future observations in features

25

MONOTONE CONSTRAINTS

Directionally consistent inputs

210 / 45 / 45

PATIENT SPLIT

Train / validation / test

49

AUTOMATED TESTS

Including temporal-leakage test

CONSTRAINT DIRECTION

Pollution, rescue use, symptoms and respiratory rate rising cannot lower constrained risk.

Improved adherence or SpO_2 cannot raise constrained risk.

Isotonic calibration fitted on validation patients only

A higher-scoring relaxed-constraint variant was rejected after failing what-if consistency checks.

PREDICTIVE PERFORMANCE

Held-out test performance

45 patients · 2,700 patient-days · synthetic event labels

ROC-AUC

0.82

95% CI 0.795–0.851

PR-AUC

0.45

CI 0.404–0.510
Base rate about 7–8%

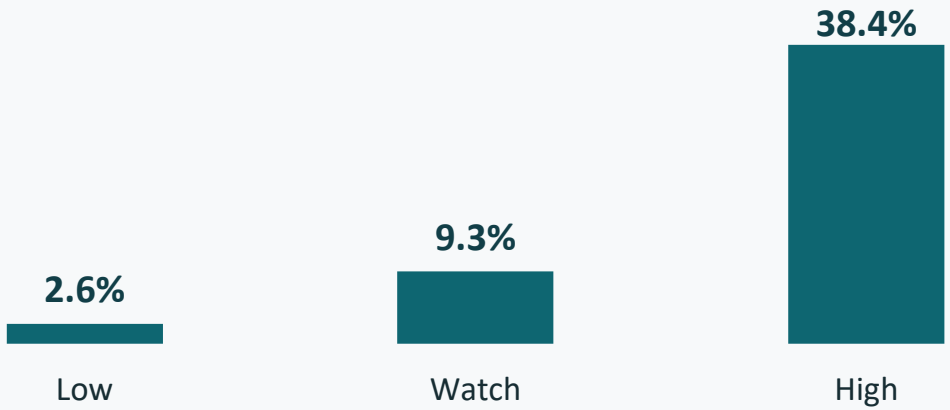
BRIER SCORE

0.0526

vs 0.0705
base-rate predictor

Observed event rate by demo risk band

These are presentation bands, not clinical thresholds.



INTERPRETATION

The pipeline learns a simulated process.

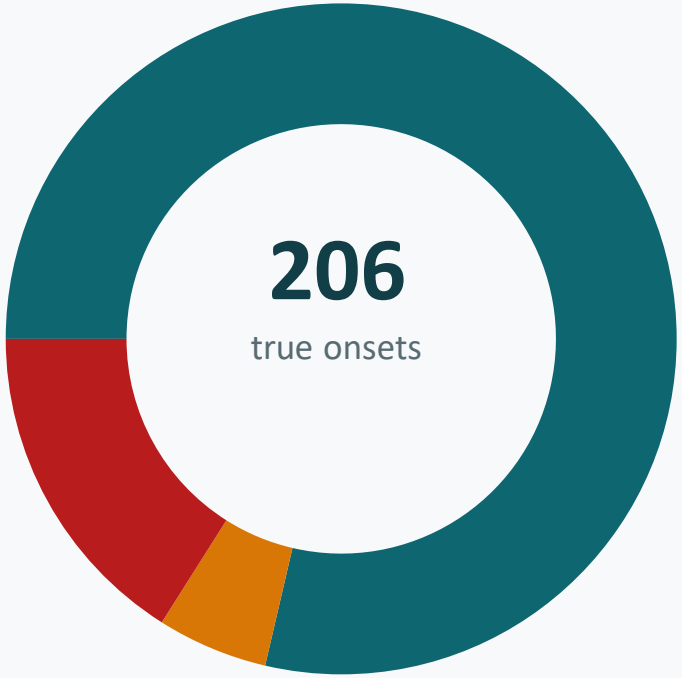
No real patient outcomes were used.
Metrics are not clinical performance.

Simulation-based results; not clinical performance.

EARLY WARNING

Onset replay reveals where the model struggles

206 true simulated onsets in the held-out test set



78.6% Anticipated · 162

5.3% Late within 6h · 11

16.0% Missed · 33

Median lead time: 12 hours

Episode type	Anticipated	Late	Missed
Non-silent	**100% (138/138)**	—	—
Silent	**35.3% (24/68)**	16.2%	48.5%

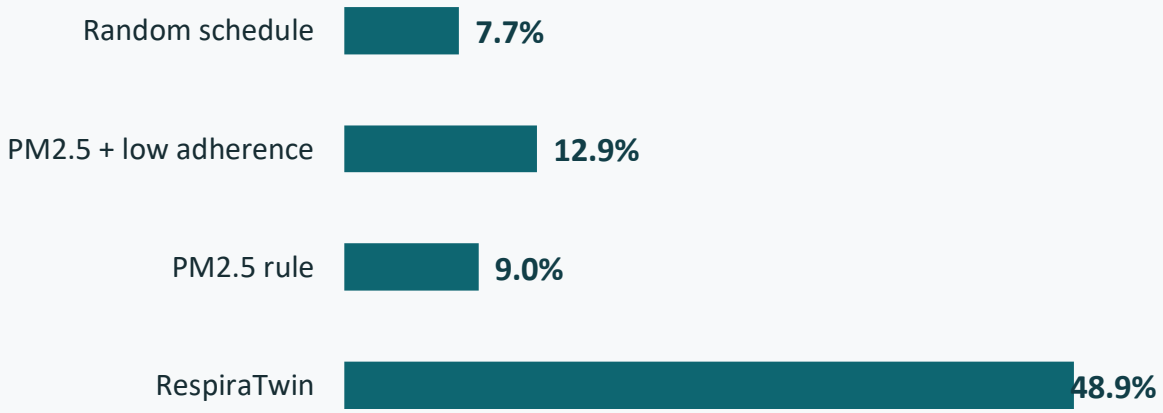
Simulation-based results; not clinical performance.

Coverage is similar to a pollution rule; alert burden differs

Comparisons use the same event and alert definitions.

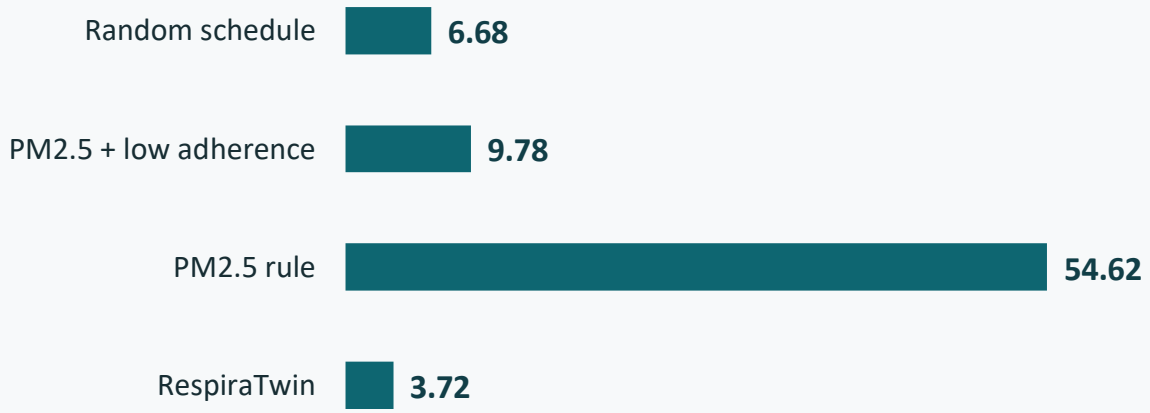
Alert precision

Share of alert episodes followed by an onset within 24h



False alert episodes per patient-month

Lower is fewer false alert episodes



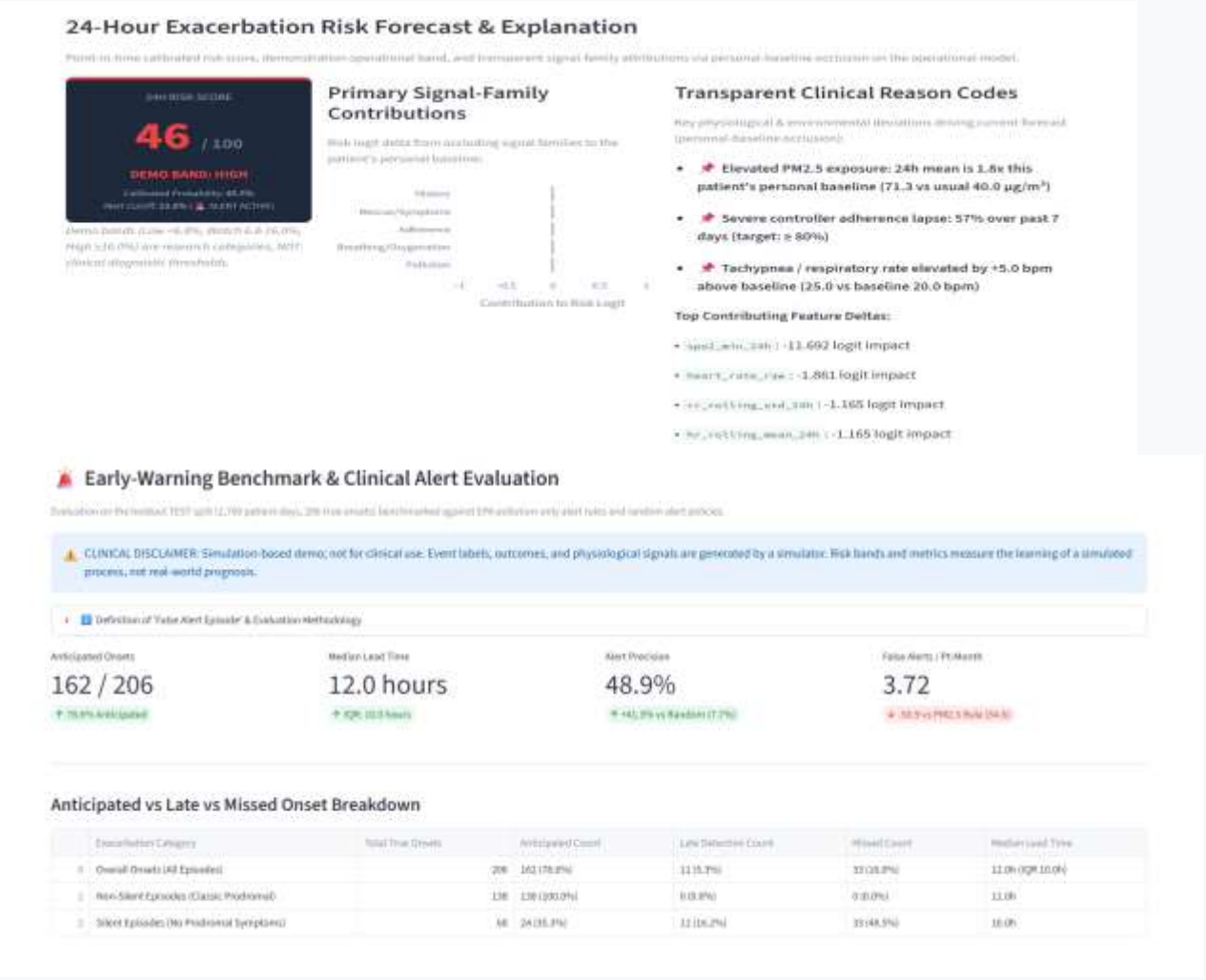
Similar coverage to a PM2.5 rule, with about 15x fewer false alerts than that rule

PM2.5 rule anticipates 83.0% of onsets vs RespiraTwin at 78.6%.

Simulation-based results; not clinical performance.

What a reviewer should see in the working app

Interface views for risk forecasting, early warning, and episode replay.



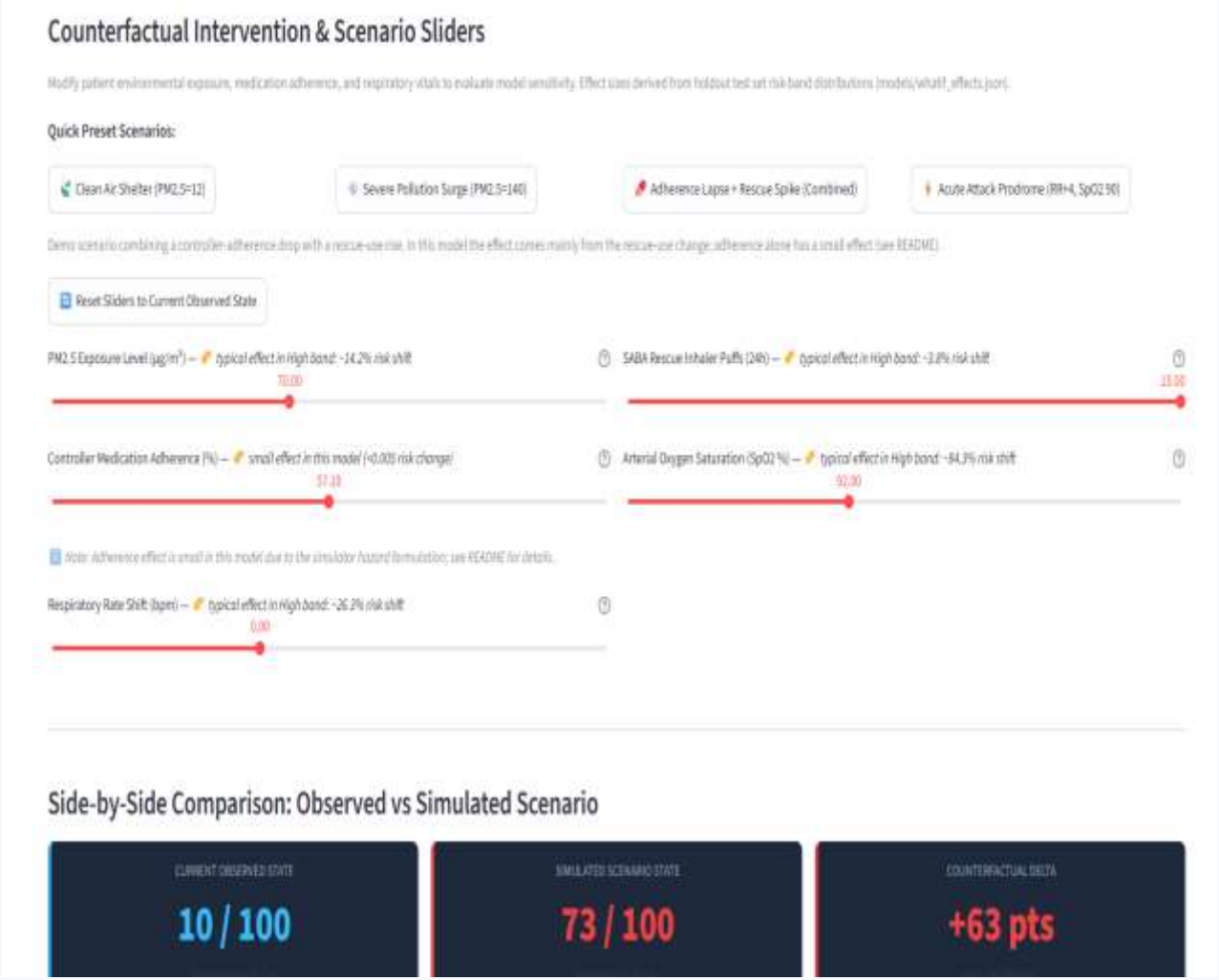
- Synthetic-patient banner
- Signal charts with provenance badges
- 0–100 risk score with Low / Watch / High demo bands
- Reason codes and twin-history timeline
- Replay of held-out episodes, including a miss

Interactive virtual patient timeline and episode replay.

WHAT-IF AND EXPLANATIONS

Explore model response without changing source history

Hypothetical scenarios are not intervention recommendations.



CONSISTENCY CHECK

100% **expected-direction results**

5 sweeps · 200 test patient-hours

- Scenario presets and sliders
- Original data never overwritten
- Personal-baseline occlusion by signal family

Caveat: adherence alone has a small effect; about 77% of test-set hours have 100% adherence and the model rarely splits on it.

A transparent proof of concept—not a clinical tool

Limits define what the results can and cannot support.

- All patient, wearable, pollution and event data are simulated.
- Metrics reflect the simulator’s assumptions.
- Silent episodes are mostly missed or caught late.
- Precision–recall trade-off changes the alert burden.
- No external or clinical validation.
- Not for clinical use; no diagnosis or treatment advice.

Synthetic data only in this prototype. Real-data use would need consent, governance, security, and applicable privacy review.

NEXT STEPS

Move from simulation to responsible evaluation

The next milestone is validation, not a stronger claim.

- Validate on real, consented, de-identified data with clinicians
- Replace synthetic pollution with sourced CPCB / OpenAQ data
- Integrate real wearable data with provenance and quality checks
- Plan prospective evaluation with appropriate governance
- Re-test what-if consistency on larger cohorts and real data

PROJECT REPOSITORY

GitHub

github.com/dhruvishah-crypto/digital-twin-challenge

TEAM CONTACT

GitHub: github.com/dhruvishah-crypto

Thank you

RespiraTwin · Research prototype

Appendix: detailed architecture

End-to-end data ingestion, rolling feature engineering, constrained forecasting, and what-if digital twin.

